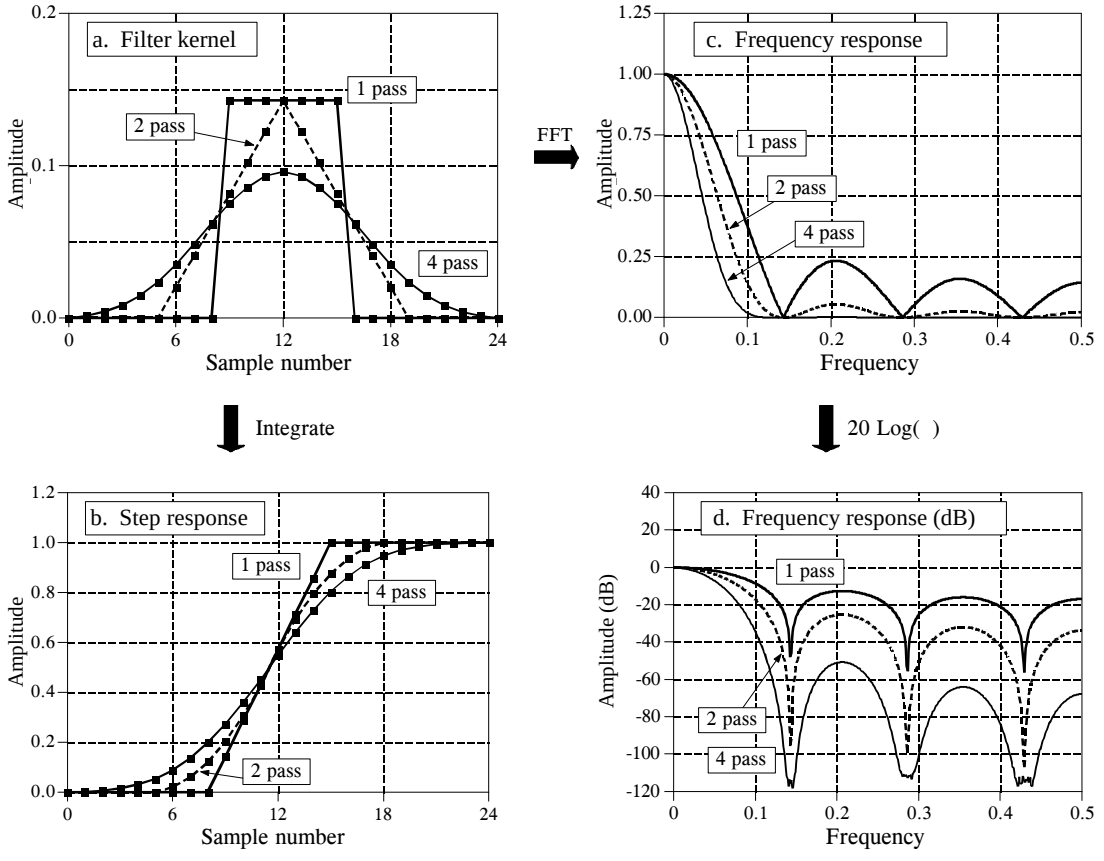

FIGURE 15-3
Characteristics of multiple-pass moving average filters. Figure (a) shows the filter kernels resulting from passing a seven point moving average filter over the data once, twice and four times. Figure (b) shows the corresponding step responses, while (c) and (d) show the corresponding frequency responses.

the signal's information is encoded in the time domain. For instance, the temperature monitor in a scientific experiment might be contaminated with 60 hertz from the power lines, 30 kHz from a switching power supply, or 1320 kHz from a local AM radio station. Relatives of the moving average filter have better frequency domain performance, and can be useful in these mixed domain applications.

Multiple-pass moving average filters involve passing the input signal through a moving average filter two or more times. Figure 15-3a shows the overall filter kernel resulting from one, two and four passes. Two passes are equivalent to using a *triangular* filter kernel (a rectangular filter kernel convolved with itself). After four or more passes, the equivalent filter kernel looks like a *Gaussian* (recall the Central Limit Theorem). As shown in (b), multiple passes produce an "s" shaped step response, as compared to the straight line of the single pass. The frequency responses in (c) and (d) are given by Eq. 15-2 *multiplied* by itself for each pass. That is, each time domain convolution results in a multiplication of the frequency spectra.